

TalkTrace Firebase Database Schema Specification

Firebase Cloud Firestore Collections & Document Field Definitions

COLLECTION 1: users (Firebase User Accounts)

Stores core account information for system users across all roles (admins, subscribers, host users). Manages unique login credentials, email verification status, and security parameters.

No	Field Name	Data Type	Key / Constraint	Description
1	uid	string	PK (Document ID)	Unique Firebase Auth User Identi...
2	name	string	NOT NULL	Full Name of User
3	email	string	UNIQUE, NOT NULL	Login Email Address
4	emailVerified	boolean	DEFAULT false	Firebase Email Verification Flag
5	role	string	DEFAULT user	Access Control Role (admin, us...
6	createdAt	timestamp	DEFAULT serverTimestamp()	Account Creation Date

COLLECTION 2: meetings (Meeting Sessions & Audio Meta)

Holds metadata for recorded meeting sessions, including language auto-detection, audio duration, processing lifecycle state, and owner association.

No	Field Name	Data Type	Key / Constraint	Description
1	id	string	PK (Document ID)	Unique Meeting UUID
2	user_id	string	FK (users.uid)	Owner User Identifier
3	bot_id	string	NULLABLE	Recall.ai Note-Taker Bot ID
4	status	string	DEFAULT processing	Overall Status (processing, co...
5	process_state	string	DEFAULT initializing	Detailed Pipeline State (trans...
6	language	string	DEFAULT auto	Google / Whisper Verified Lang...
7	duration_seconds	number	DEFAULT 0	Audio Duration in Seconds
8	created_at	number (epoch)	NOT NULL	Session Epoch Timestamp (ms)

COLLECTION 3: meeting_keys (Zero-Trust Cryptographic Key Vault)

Encrypted vault storing AES-256 keys and initialization vectors (IV) for local zero-trust audio and transcript encryption.

No	Field Name	Data Type	Key / Constraint	Description
1	meeting_id	string	PK (Document ID), FK	Associated Meeting ID
2	encrypted_key_blob	string	NOT NULL	Protected Key Vault Blob (AES/...

COLLECTION 4: transcript_revisions (Version History & Hash Audit Log)

Tracks version history and SHA-256 cryptographic hashes for transcripts and executive summaries to guarantee audit immutability.

No	Field Name	Data Type	Key / Constraint	Description
1	id	string	PK (Document ID)	Revision Record ID
2	meeting_id	string	FK (meetings.id)	Associated Meeting ID
3	version	number	NOT NULL	Revision Version Number (1, 2,...
4	type	string	NOT NULL	Revision Artifact Type (transc...
5	content_hash	string	NOT NULL	SHA-256 Cryptographic Hash
6	file_path	string	NOT NULL	Encrypted Vault File Path
7	edited_at	number (epoch)	NOT NULL	Revision Edit Timestamp (ms)

COLLECTION 5: speaker_profiles (Recurring Speaker Voice & Name Registry)

Stores persistent participant voice profiles and custom name mappings across meetings for automated Pyannote speaker recognition.

No	Field Name	Data Type	Key / Constraint	Description
1	id	string	PK (Document ID)	Profile Record ID
2	user_id	string	FK (users.uid)	Owner User ID
3	speaker_label	string	NOT NULL	Pyannote Diarization Label
4	speaker_name	string	NOT NULL	Real Human Name (e.g. Abin Geo...
5	voice_signature	string	NULLABLE	Dense Voice Embedding Vector
6	updated_at	number (epoch)	NOT NULL	Last Update Timestamp (ms)

COLLECTION 6: action_items (AI Action Items & Task Manager Board)

Contains structured tasks automatically extracted by local LLMs (Ollama) from meeting transcripts, with priorities.

No	Field Name	Data Type	Key / Constraint	Description
1	id	string	PK (Document ID)	Action Item Unique ID
2	meeting_id	string	FK (meetings.id)	Originating Meeting ID
3	task	string	NOT NULL	Action Item Task Description
4	assignee	string	DEFAULT Unassigned	Responsible Participant
5	deadline	string	DEFAULT ASAP	Targeted Due Date / Timeframe
6	priority	string	DEFAULT Medium	Task Urgency Rating (High, Med...
7	completed	boolean	DEFAULT false	Task Completion Status
8	confidence	number	DEFAULT 0.95	LLM Extraction Confidence Score

COLLECTION 7: calendar_events (Google & Outlook Synced Meetings)

Stores synced calendar events from Google Calendar and Microsoft Outlook Graph APIs for automated bot scheduling.

No	Field Name	Data Type	Key / Constraint	Description
1	id	string	PK (Document ID)	Calendar Event Unique ID
2	user_id	string	FK (users.uid)	Associated User Account
3	title	string	NOT NULL	Meeting Title / Subject
4	start_time	timestamp	NOT NULL	Scheduled Start DateTime
5	meeting_link	string	NOT NULL	Google Meet / Teams / Zoom URL
6	source	string	NOT NULL	Calendar Source API (Google, O...
7	bot_scheduled	boolean	DEFAULT false	Auto-Join Bot Scheduled Flag

COLLECTION 8: search_history (Natural Language Vector RAG Query Logs)

Stores natural language search queries and 128-dimensional sentence embedding vector logs for client-side search audit.

No	Field Name	Data Type	Key / Constraint	Description
1	id	string	PK (Document ID)	Log ID
2	user_id	string	FK (users.uid)	Query User ID
3	query	string	NOT NULL	Natural Language Search String
4	created_at	number (epoch)	NOT NULL	Search Timestamp (ms)